

UPPSALA UNIVERSITET

- What is AI and ML?
- Course information
- Introduction to Supervised Learning
  - Example: Logistic regression
- Optimization in Machine Learning

Machine learning – Block 1(a)

Måns Magnusson
Department of Statistics, Uppsala University

Autumn 2026



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This block

- What is AI and Machine Learning?
- Course Information and Practicalities
- Introduction to Supervised Learning
- (Stochastic) Gradient Descent
- Regularization



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Section 1

What is AI and ML?



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What exactly is machine learning and artificial intelligence?

The word "AI" is often used quite loosely:

To briefly explain how Linear Regression helped us reverse engineer the BSR equation, let's break it down. Linear Regression is an AI equation that finds the proper coefficients for an equation by sorting through massive amounts of data. The equation looks something like BSR = X(a)+ Y(b) + Z(c)..... and so and and so forth.



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What is Artificial Intelligence?

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What is Artificial General Intelligence?

Artificial general intelligence (AGI) is the hypothetical intelligence of a machine that has the capacity to understand or learn any intellectual task that a human being can. – Wikipedia

Also called:

1. Strong AI
2. General AI
3. Full AI



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Artificial super intelligence (ASI) is "any intellect that greatly exceeds the cognitive performance of humans in virtually all domains of interest" – Nick Bostrom



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A Definition of AGI (Hendrycks et al., 2025)

- One proposed operational definition: matching the cognitive versatility and proficiency of a well-educated adult.
- Cattell–Horn–Carroll (CHC) theory of human cognition.
- Evaluates ten broad cognitive domains:
  - Knowledge, Reading & Writing, Math, Reasoning
  - Working/Long-term Memory, Visual, Auditory, Speed
- Reported framework scores: GPT-4: 27%, GPT-5: 57%.
- These scores depend on this framework; they are not percentages of human intelligence.

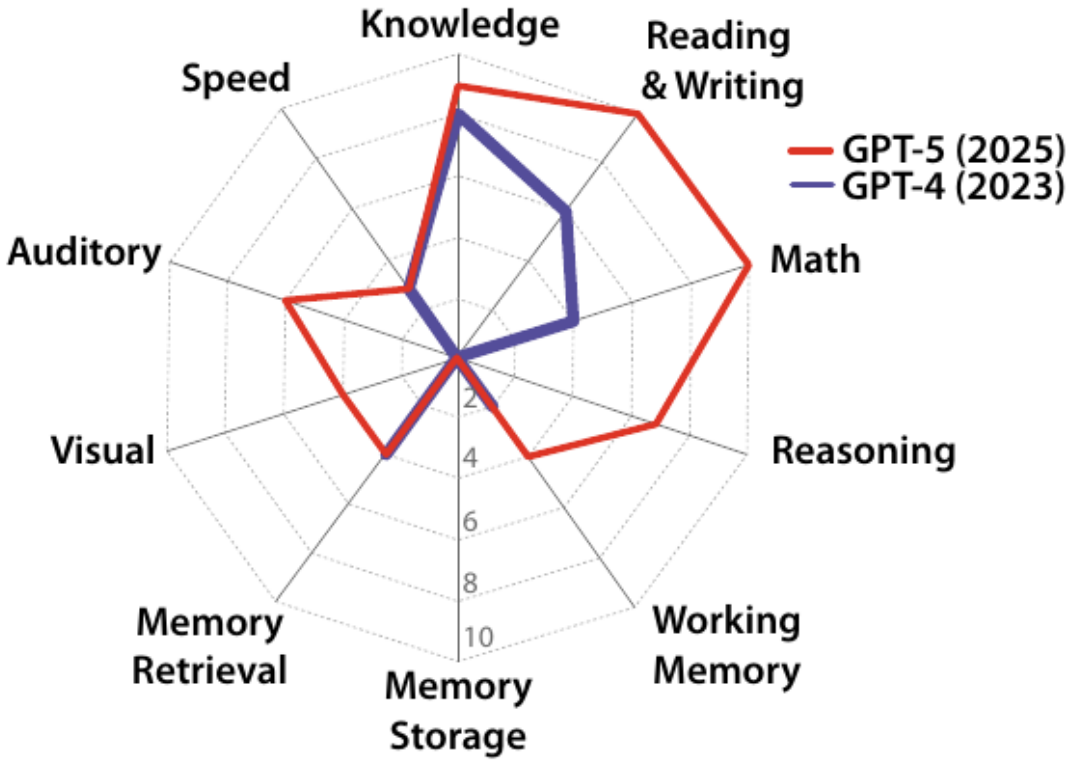
Hendrycks et al. (2025), arXiv v3, 3 December 2025.



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Cognitive profiles under the proposed framework (Hendrycks et al., 2025).
See the versioned paper on the preceding slide for the reported scores.



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What is Machine Learning?

Machine Learning is the field of study that gives the computer the ability to learn without being explicitly programmed. – Arthur Samuel (1959)

A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E. – Tom Mitchell (1998)



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Learning from data. – Hastie, Tibshirani, Friedman (2009)

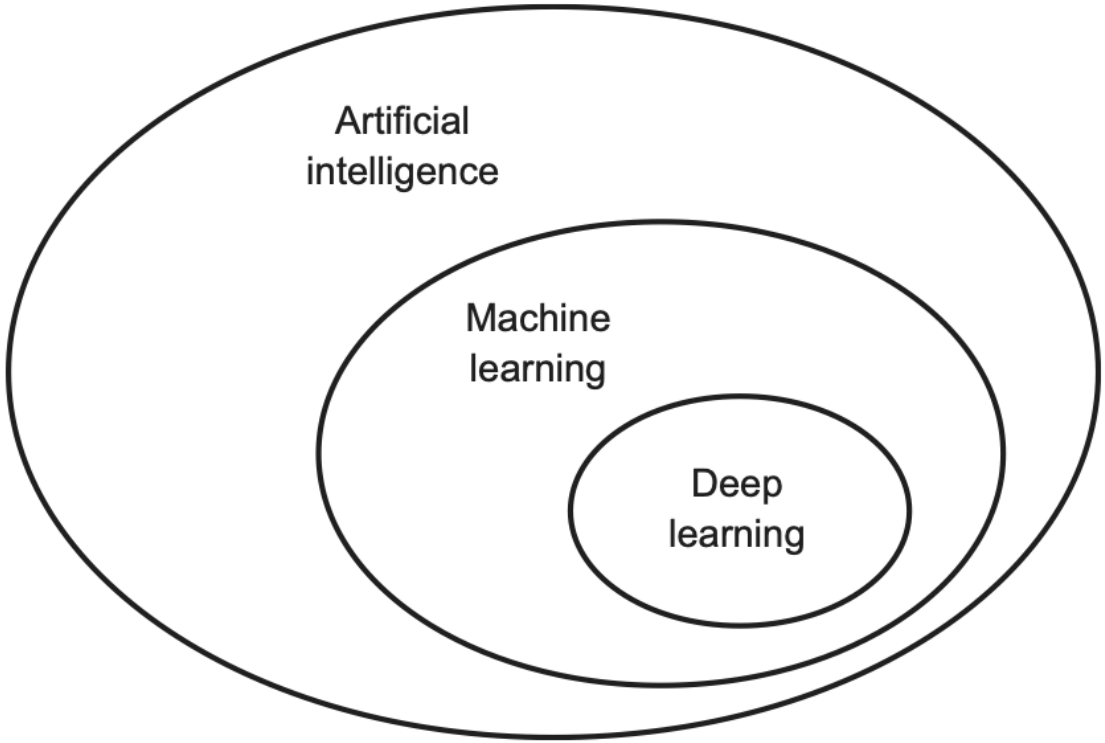


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What is Machine Learning?

Figure: ML, AI and DL (Chollet, 2018, Figure 1.1)



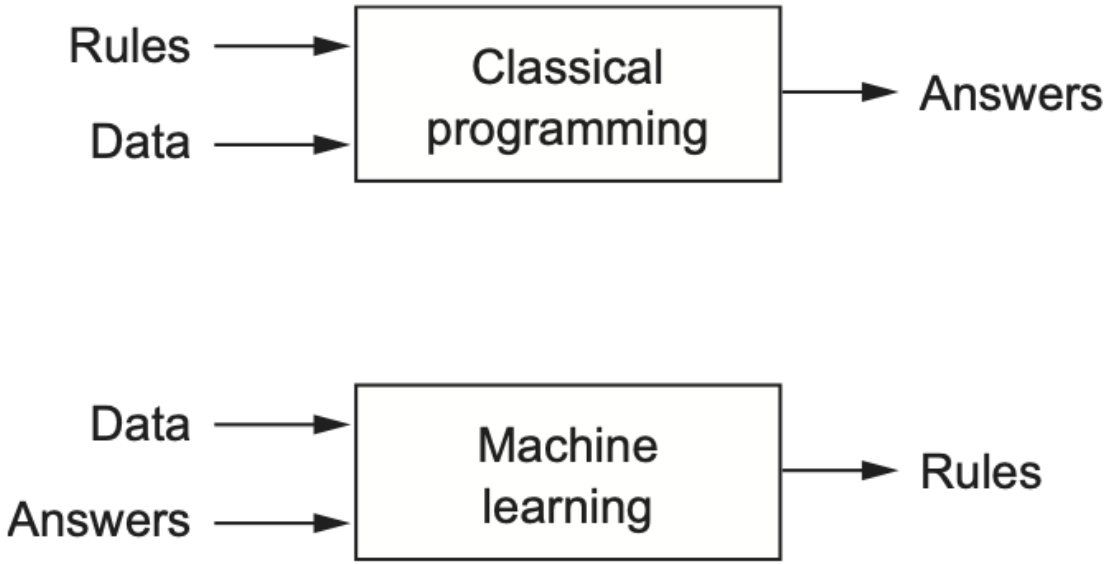


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Computer Science and Machine Learning

Figure: A new paradigm? (Chollet, 2018, Figure 1.2)





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Statistics and Machine Learning

Figure: Regression vs. Pure Predictions (Efron, 2020, Table 5)

Table 5. A comparison checklist of differences between traditional regression methods and pure prediction algorithms.

	Traditional regressions methods	Pure prediction algorithms
1.	Surface plus noise models (continuous, smooth)	Direct prediction (possibly discrete, jagged)
2.	Scientific truth (long-term)	Empirical prediction accuracy (possibly short-term)
3.	Parametric modeling (causality)	Nonparametric (black box)
4.	Parsimonious modeling (researchers choose covariates)	Anti-parsimony (algorithm chooses predictors)
5.	x $p \times n$: with $p \ll n$ (homogeneous data)	$p \gg n$, both possibly enormous (mixed data)
6.	Theory of optimal inference (mle, Neyman–Pearson)	Training/test paradigm (Common Task Framework)



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Related terminology across communities

- Machine learning has developed in parallel with Statistics
- Terms often overlap, but are not always equivalent:
  1. Time series regression (ML) vs. Scalar-on-function regression (Stats)



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  3. Learning (ML) overlaps with estimation (Stats)



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  3. Learning (ML) overlaps with estimation (Stats)
  4. Weights (ML) vs. Parameters (Stats)



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  3. Learning (ML) overlaps with estimation (Stats)
  4. Weights (ML) vs. Parameters (Stats)
  5. Features (ML) vs. Covariates (Stats)



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Different flavors of ML

- Supervised learning
- Unsupervised learning
- Reinforcement learning



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The Bitter Lesson (Rich Sutton, 2019)

- Sutton's argument: general methods that exploit computation have repeatedly outperformed approaches built around hand-coded domain knowledge.



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- *Search and learning* can exploit growing computational resources to improve performance.



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- Hand-coded knowledge can help initially, but may limit how much a method benefits from additional computation.
- *Search and learning* can exploit growing computational resources to improve performance.
- Examples: Chess, Go, speech recognition, vision, language models — all advanced by scaling compute, not by encoding expert knowledge.



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- Hand-coded knowledge can help initially, but may limit how much a method benefits from additional computation.
- *Search and learning* can exploit growing computational resources to improve performance.
- Examples: Chess, Go, speech recognition, vision, language models — all advanced by scaling compute, not by encoding expert knowledge.
- Takeaway: favour methods that scale with computation. This does not rule out useful architectures, inductive biases, or domain knowledge.



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Questions?

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Section 2

Course information



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Course information

The aims of this course are that you should:



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Course information

The aims of this course are that you should:

1. get a good knowledge of a large number of machine learning models,
2. become able to use methods for evaluating and improving predictive models,
3. become able to handle big data,
4. become able to train and use machine learning models in R,
5. become able to train and use neural networks using Keras/TensorFlow.
6. become able to describe and discuss ethical aspects of big data and black box-models,



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Course Outline

Two main parts:

- Core Content (8 lecture blocks):
  - Supervised learning (5 blocks)
    - Introduction, statistical learning (1 block)
    - Tree-based methods (1 block)
    - Neural Networks (3 block)
  - Unsupervised learning (2 blocks)
  - Reinforcement learning (1 block)



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- Oral exams for all eight assignments



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- Mini-project on a supervised project (2-3 students)



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Exact dates and details; see the course page.



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Core Content

- Each block consist of:
  - Online video material (optional)
  - Reading assignments (approx. 2-4h, 50-90 pages a week)
  - One-two Lecture(s) (optional)
  - An individual computer assignment (approx. 14-16h).
  - Zoom question hours (optional)



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  - Zoom question hours (optional)
- Reading: Mandatory and optional (overlap)
- Recommended workflow for each block
  - Do the reading assignments
  - Watch the videos (optional)
  - Attend the lecture (optional) to ask questions.
  - Do the computer assignment
  - Attend the zoom lab session (optional) to ask questions.



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- Present overall theory, concepts and content (overview)



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- Guest lectures on the course (worthwhile):
  1. Jonas Wallin, Lund University (regularization)



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- See TimeEdit for lecture dates and times.



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Examination

- To pass (G): All labs (at 75%), mini-project and oral exam needs to be passed



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- To pass with distinction (VG): 7/10 VG points



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Examination

- To pass (G): All labs (at 75%), mini-project and oral exam needs to be passed
- To pass with distinction (VG): 7/10 VG points
- Each assignment has an extra (VG) task worth 1 VG point.
- The mini-project is worth 2 VG-points (if it is passed with distinction).
- Ph.D. students: I suggest you get VG to pass the course. Make the project a potential paper.
- Reassessment of grades (supply form to course admin)
- Failing the course: You will need to redo all assignments, oral exam and mini-project.



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Computer Assignments

- Main part of the course
  Learning by doing
- Machine learning = Statistics + Computer Science
  Hence a lot of programming



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- We will mark and return each assignment within 10 working days.



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- Zoom sessions:
  1. First lab each week will include a 15 min introduction
  2. Our focus: Help during computer labs - less focus on written feedback
  3. Ask questions! This is your time.



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- What is AI and ML?
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Computer Assignments

- Don't write your name anywhere!
- Do the assignment evaluation
- *Important!* Don't share your solutions with other students. But feel free to discuss concepts and approaches!
- Zoom sessions:
  1. First lab each week will include a 15 min introduction
  2. Our focus: Help during computer labs - less focus on written feedback
  3. Ask questions! This is your time.



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Learning with LLMs

- Discuss ideas with LLMs as with fellow students: ask for explanations and get feedback to support your learning.
- Optional study support: first upload `AI_TUTOR.md` and ask the tool to follow it; then upload the assignment instructions.
- Ask for explanations, hints, and separate practice examples. Your submitted code, calculations, and analysis must be your own.
- Do not upload your own report, answers, or assessed code. Be ready to explain every part at the oral exam.
- No LLM help with your original oral question and answer. Explicit LLM experiments keep their stated permissions.
- You remain responsible for the course rules, even if the tutor offers a completed solution.

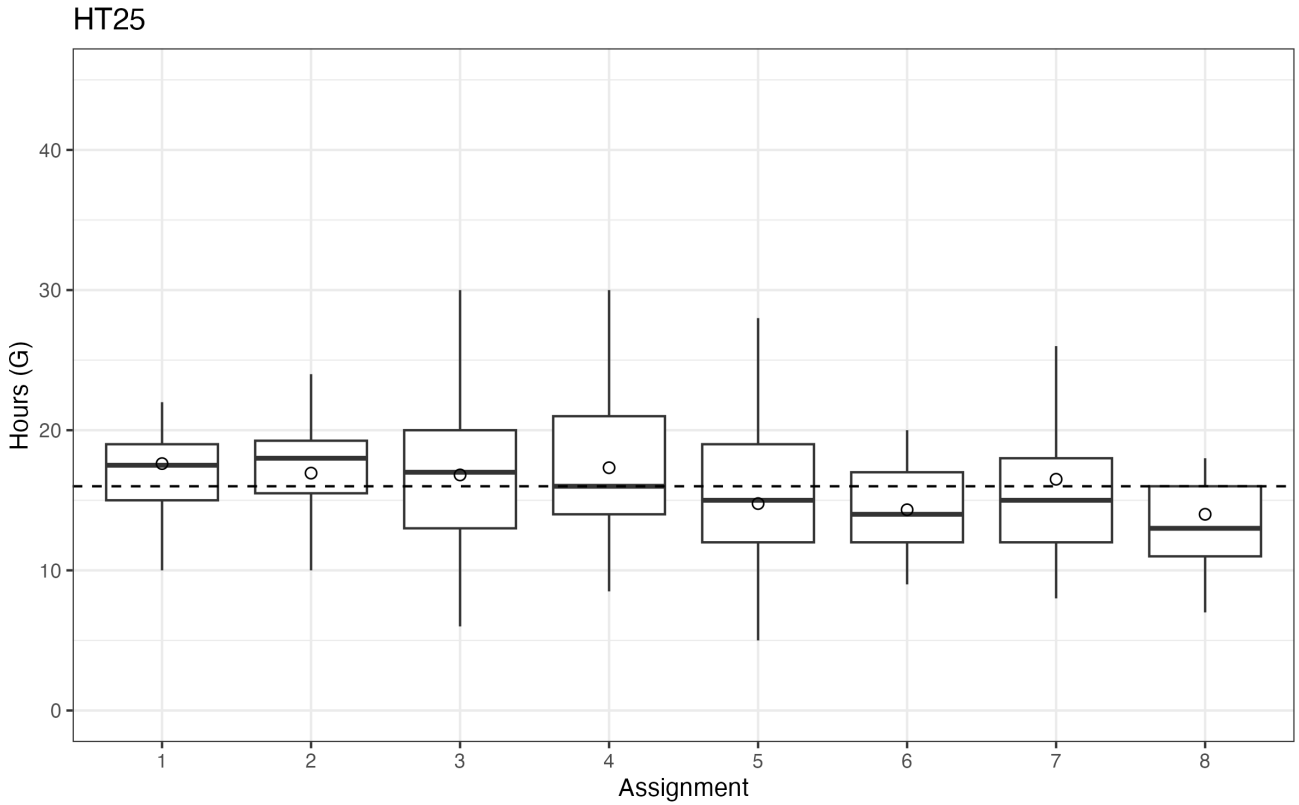


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Computer Assignments

Figure: Workload last year to pass (G)



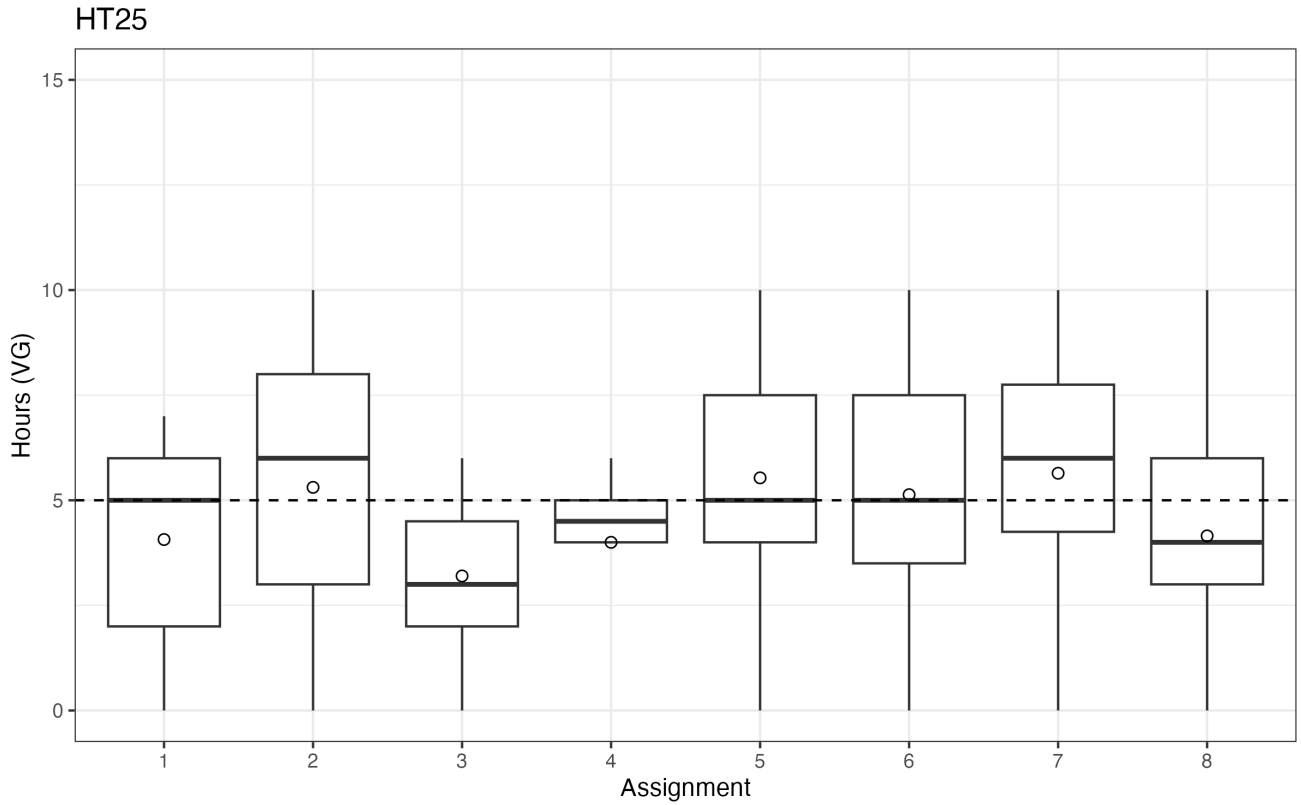


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Figure: Workload last year to pass with distinction (VG)





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Oral Exam

- Weekly opportunities from week 3. Choose how often to attend; see Studium/TimeEdit for dates and rooms.
- Take 1–3 assignments per session. Submit them in Studium one week before the exam. Written grading may still be pending.
- One randomly drawn question per assignment, with pot. follow-up questions. Each assignment is assessed separately.
- Explain your own code, results, and reasoning, including parts beyond the published questions. You should not need to memorise code or formulas! Let us know if you feel you need to.
- You need to pass the first time to get the VG point.
- Prepare with the oral exam instructions and question bank for all eight assignments and more detailed information.



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Mini-project

- See project instructions on webpage for details.



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- Supervised problem of choice on real data.
- 2-3 students.



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- Supply step 2 proposal at the end of block 6.
- Project will last two weeks (half time) - but start earlier. Good case to show potential employers.
- Recommended data: Images, text or tabular data (e.g. avoid time series).
- Feel free to build upon the Bayesian project (e.g. compare with Bayesian methods).
- Approximate 40 hours of work *per student*.



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- Project oral presentation (10-15 minutes)



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- The project should result in a 4 page report (PDF) using the ICML LaTeX template (see course page).
- Project oral presentation (10-15 minutes)
- The first author is corresponding author
- Mini-project and master thesis:
  - The mini-project can be used to explore your thesis project data



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Practicalities

- Course page: GitHub – corrections and PRs welcome!



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  - Bishop & Bishop (2024) *Deep Learning - Foundations and Concepts*.
  - Hastie, Tibshirani & Friedman (2009). *Elements of Statistical Learning*.
  - Sutton and Barto (2020) *Reinforcement learning: An introduction*
  - Chollet & Allaire (2018) *Deep Learning with R* (optional)
  - Additional articles, tutorials, videos etc. posted on course (github) homepage
  - Reading priorities: see the block reading instructions.



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  - Additional articles, tutorials, videos etc. posted on course (github) homepage
  - Reading priorities: see the block reading instructions.
- If you have complaint - reach out to me in assignment evaluations!



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Course improvements since last year

- More automatic feedback: expanded `markmyassignment` (MMA) checks for your R functions.
- More transparent grading: grading rubrics with clearer assessment criteria and point allocation.
- All assignments revised: clearer tasks, worked examples, hints, and checks of your results.
- Updated software support: Keras 3 examples and setup instructions for local R and Google Colab.



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Your feedback: learning and exams

- LLMs as study partners: load `AI_TUTOR.md`, then the assignment brief. Get hints; submit your own work.
- Flexible oral exams: weekly from week 3; choose when to take 1–3 submitted assignments per session.
- Trying seminars: optional small-group practice with difficult oral questions and short derivations; we will evaluate how it works.
- Clearer reading guidance: closer alignment with Bishop & Bishop and self-contained explanations of examples adapted from Goodfellow.



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Questions?



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Section 3

Introduction to Supervised Learning



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Supervised learning

Figure: Relationship between apartment size and price (source)



Problem: We want to predict the price of a new apartment.



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Pair discussion: apartment prices

We want to predict an apartment's final selling price when it is first listed for sale.

Pairs, 1–2 minutes:

1. What is the outcome, and what are two usable predictors?
2. Name one tempting predictor that would be unavailable at prediction time. Why would using it mislead us?



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Supervised learning

- General problem: We have *training* data

$$\mathbf{d} = \{(y_i, \mathbf{x}_i), i = 1, ..., n\}\,.$$

- $\mathbf{x}_i$ = features/input/predictors/features/independent variables
- y_i = labels/output/dependent variable
- We want to *learn* a function $\hat{f}$ giving predictions $\hat{y}_{\text{new}} = \hat{f}(\mathbf{x}_{\text{new}})$ with as good performance as possible.



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- We want to *learn* a function $\hat{f}$ giving predictions $\hat{y}_{\text{new}} = \hat{f}(\mathbf{x}_{\text{new}})$ with as good performance as possible.
- Regression problems: $y_i \in \mathbb{R}$
- Classification problems: $y_i \in a, b, c, ...$ where $a, b, c...$ are discrete classes.



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Example of supervised problems

Any examples of applications?



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Example of supervised problems

Any examples of applications?

- Is this e-mail message spam or not?



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Example of supervised problems

Any examples of applications?

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- Image recognition/classification



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Any examples of applications?

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- Image object tracking (position in a video)



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- Does this fingerprint belong to an employee or not?



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- Face recognition



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- Is this tumour malign or not?



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Logistic regression and classification

When the y_i in a regression problem is binary (or more generally, categorical), it becomes a classification problem.



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Logistic regression and classification

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The question that the model tries to answer is: does this observation belong to class 0 or class 1?



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Logistic regression and classification

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Logistic regression is a workhorse for classification problems.



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Logistic regression

When analysing binary data $y_1, \ldots, y_N$, we usually assume that the Y_i follow binomial (or Bernoulli) distributions.



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Logistic regression

When analysing binary data $y_1, \ldots, y_N$, we usually assume that the Y_i follow binomial (or Bernoulli) distributions.

Given the predictors, assume independent responses with $Y_i \mid \mathbf{x}_i \sim \mathit{Bernoulli}(\pi_i)$.



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Given the predictors, assume independent responses with $Y_i \mid \mathbf{x}_i \sim \mathit{Bernoulli}(\pi_i)$.

$Y_i \in \{0, 1\}$ with $\pi_i = P(Y_i = 1 \mid \mathbf{x}_i) = E(Y_i \mid \mathbf{x}_i)$.



Logistic regression

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$Y_i \in \{0, 1\}$ with $\pi_i = P(Y_i = 1 \mid \mathbf{x}_i) = E(Y_i \mid \mathbf{x}_i)$.

- The natural parameter of the binomial distribution is

$$g(\pi_i) = \log\left(\frac{\pi_i}{1 - \pi_i}\right),$$

  called the logit or log odds.



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  called the logit or log odds.

- A GLM using this link function is called logistic regression; other links are possible for binary responses.



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Logistic regression

There are two equivalent formulas for logistic regression:

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}, \qquad i = 1, \ldots, N$$

and

$$\pi_i = \frac{\exp\left(\beta_0 + \sum_{j=1}^{p} \beta_j x_{ij}\right)}{1 + \exp\left(\beta_0 + \sum_{j=1}^{p} \beta_j x_{ij}\right)}.$$



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Logistic regression: Prediction

- We *train* a logistic regression model by MLE to estimate $\hat{\theta}$.
- With perfect separation, a finite unregularized MLE may not exist; regularization can help.
- For a new observation, compute $\hat{p}_{\text{new}} = g^{-1}(\mathbf{x}_{\text{new}}^T \hat{\theta})$ (including an intercept).
- We use a decision rule to predict value 0 or 1:

$$\hat{y}_{\text{new}} = \begin{cases} 1, & \text{if } \hat{p}_{\text{new}} \geq 0.5 \\ 0, & \text{otherwise} \end{cases}$$

- Threshold 0.5 minimizes expected 0–1 loss when probabilities are correct; unequal error costs can justify another threshold.



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Pair discussion: probabilities and decisions

Two emails have predicted spam probabilities 0.6 and 0.9. We classify an email as spam if its predicted probability is at least 0.5.

Pairs, 1–2 minutes:

1. What information is lost if we report only the predicted classes?
2. Suppose discarding a legitimate email is much more costly than letting spam through. Would you raise or lower the threshold? Why?

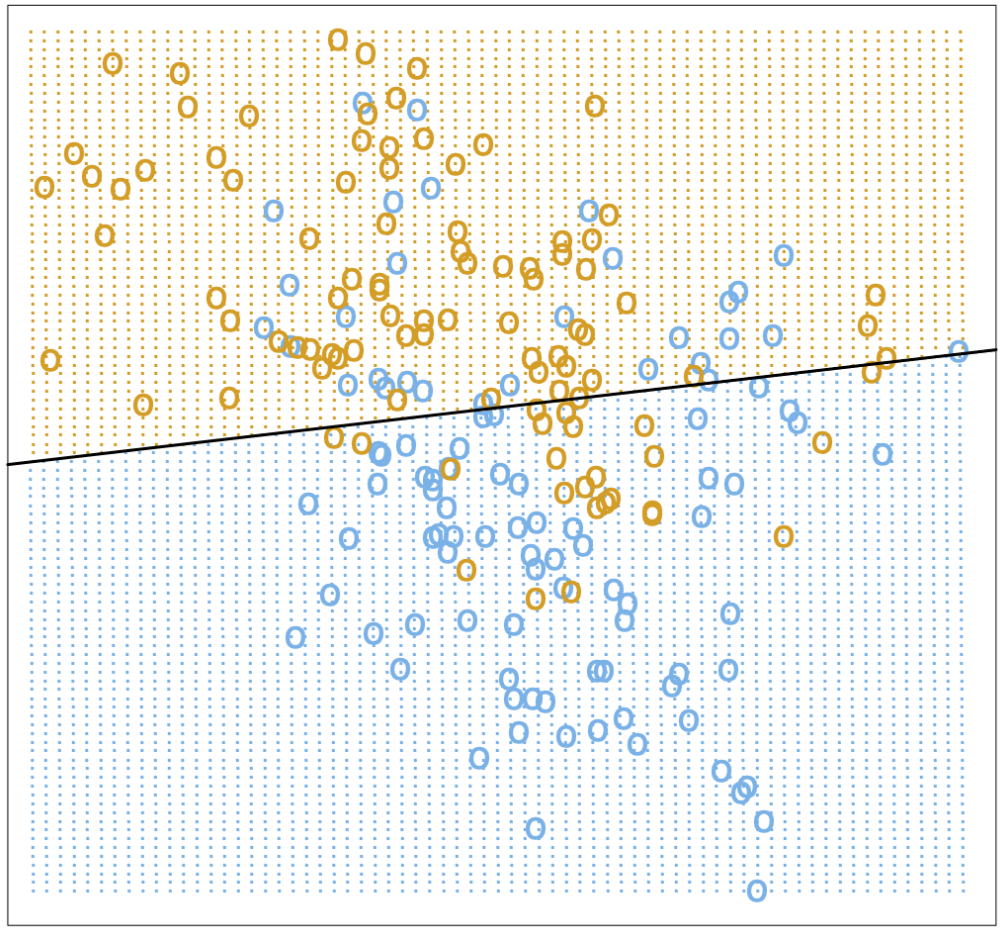


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Logistic regression: Example

Figure: Decision boundary with two covariates (Hastie et al, 2009, Figure 2.1)





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An example: E-mail classification

E-mail Spam

Predict if a new, unseen, e-mail is spam or ham.





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Questions?

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Section 4

Optimization in Machine Learning



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Training of ML algorithms

1. Training is usually done by minimizing the objective/loss/cost function $L(\theta)$ for $\theta \in \mathbf{R}^P$.
2. Example: Logistic regression, here we can use the negative mean log-likelihood as loss function:

$$L(\theta, \mathbf{y}, \mathbf{X}) = -\frac{1}{N} \log \prod_{i=1}^{N} p_i^{y_i} (1 - p_i)^{1-y_i} ,$$

where

$$\log \frac{p_i}{1 - p_i} = \mathbf{x}_i^T \theta ,$$



Training of ML algorithms

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where

$$\log \frac{p_i}{1 - p_i} = \mathbf{x}_i^T \theta ,$$

3. In Machine Learning: P and N might be very large...



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Gradient Descent

1. The workhorse of Machine Learning

$$\theta_t = \theta_{t-1} - \eta \nabla L(\theta_{t-1}, \mathbf{X}, \mathbf{y}),$$

where

$$\nabla f(p) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(p) \\ \vdots \\ \frac{\partial f}{\partial x_n}(p) \end{bmatrix}$$

2. ∇L is the full training-objective gradient (no sampling).
3. $L(\theta)$ needs to be differentiable



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Pair discussion: choosing a step size

Consider $L(\theta) = \theta^2$, starting at $\theta_0 = 1$.

One update is $\theta_1 = \theta_0 - \eta L\prime(\theta_0)$, with $L\prime(\theta) = 2\theta$.

Pairs, 1–2 minutes:

1. Calculate θ_1 for $\eta = 0.1$ and $\eta = 1.1$.
2. Compare $L(\theta_1)$ with $L(\theta_0)$. Does moving against the gradient always reduce the loss?



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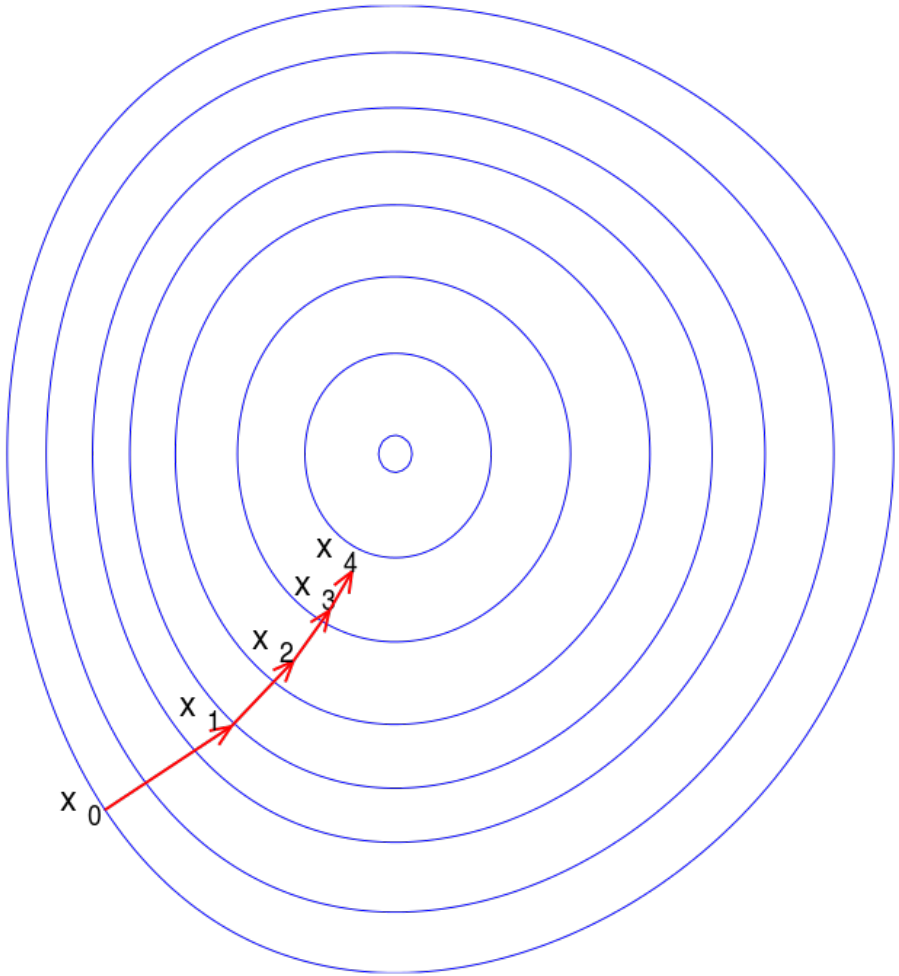
Gradient Descent Analogy

Figure: Gradient Descent Analogy (source)



Gradient Descent (cont.)

Figure: Gradient Descent (source)





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Why Gradient Descent?

- Gradient Descent is a poor algorithm (Newtons method, Iteratively Reweighted Least Squares are 'better')
- So why is gradient descent relevant?



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Why Gradient Descent?

- Gradient Descent is a poor algorithm (Newtons method, Iteratively Reweighted Least Squares are 'better')
- So why is gradient descent relevant?
- Two benefits of gradient descent:
  1. Only uses the gradient—scales to large P
  2. Can scale to large data with Stochastic Gradient Descent—scales to large N



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Stochastic Gradient Descent

- Use the mean observation loss: $L(\theta) = \frac{1}{N} \sum_{i=1}^{N} \ell_i(\theta)$.



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Stochastic Gradient Descent

- Use the mean observation loss: $L(\theta) = \frac{1}{N} \sum_{i=1}^{N} \ell_i(\theta)$.
- Let ℓ_i be observation i's loss. Sample i_t uniformly from $\{1, \ldots, N\}$:

$$\widehat{\nabla L}_t(\theta) = \nabla \ell_{i_t}(\theta), \qquad E_{i_t}[\widehat{\nabla L}_t(\theta)] = \nabla L(\theta)\,,$$

The hat denotes a sampling estimate of the full gradient. Each observation gradient is computed exactly.



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Stochastic Gradient Descent

- This gives the following algorithm:

$$\theta_t = \theta_{t-1} - \eta_t \widehat{\nabla L}_t(\theta_{t-1}),$$

  where $\widehat{\nabla L}_t$ uses a uniformly sampled observation at each update.

- *Note!*
  We need to have an unbiased estimator for $\nabla L(\theta, \mathbf{X}, \mathbf{y})$



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- *Note!*
  We need to have an unbiased estimator for $\nabla L(\theta, \mathbf{X}, \mathbf{y})$
- What is an iteration?
- Epochs vs. Iterations



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Stochastic Gradient Descent

- Learning rate η_t matters; constant rates are also used in finite training.
- Will it converge to an optimum?



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- For asymptotic convergence with persistent gradient noise, use decreasing step sizes.



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Stochastic Gradient Descent

- Learning rate η_t matters; constant rates are also used in finite training.
- Will it converge to an optimum?
- For asymptotic convergence with persistent gradient noise, use decreasing step sizes.
- Robbins–Monro (1951) step-size conditions:
  1. $\eta_t \geq 0 \; \forall t \geq 0$
  2. $\sum_t^\infty \eta_t = \infty$
  3. $\sum_t^\infty \eta_t^2 < \infty$



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  One choice satisfying these conditions is:

$$\eta_t = \frac{1}{t}, \qquad (t \geq 1)$$

- Also require regularity, controlled gradient noise, and stable iterates; nonconvex losses need not reach a global optimum.



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Mini-batch gradient descent

- Can we estimate the gradient in a better way?



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Mini-batch gradient descent

- Can we estimate the gradient in a better way?
- Take a uniformly sampled batch S_t of B indices:

$$\widehat{\nabla L}_t(\theta) = \frac{1}{B} \sum_{i \in S_t} \nabla \ell_i(\theta), \qquad E_{S_t}[\widehat{\nabla L}_t(\theta)] = \nabla L(\theta).$$

$$\theta_t = \theta_{t-1} - \eta_t \widehat{\nabla L}_t(\theta_{t-1}).$$

Sample without replacement within each fresh batch. In practice, one often shuffles data and processes batches once per epoch.



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  Sample without replacement within each fresh batch. In practice, one often shuffles data and processes batches once per epoch.
- B is usually set to optimize hardware



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Pair discussion: iterations and epochs

We have 1,000 training observations and mini-batches of 100. Each epoch visits every observation once; each batch gives one parameter update.

Pairs, 1–2 minutes:

1. How many updates are made in one epoch?
2. Why do these updates generally differ from one full-data gradient update, even if we choose the learning rates carefully?



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SGD with momentum

- SGD can be slow to converge due to 'jumping' behaviour



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SGD with momentum

- SGD can be slow to converge due to 'jumping' behaviour
- Momentum accumulates exponentially weighted, learning-rate-scaled gradients. Aligned directions reinforce movement.



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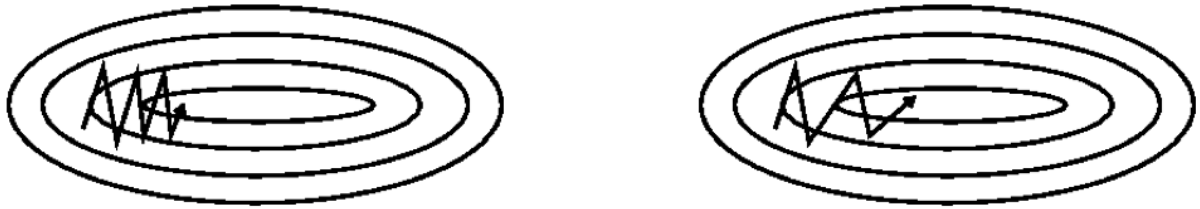
SGD with momentum

- SGD can be slow to converge due to 'jumping' behaviour
- Momentum accumulates exponentially weighted, learning-rate-scaled gradients. Aligned directions reinforce movement.
- $0 \leq \alpha < 1$ controls momentum; initialize $m_0 = 0$.

$$m_t = \alpha m_{t-1} + \eta_t \widehat{\nabla} L_t(\theta_{t-1}),$$

$$\theta_t = \theta_{t-1} - m_t,$$

Figure: SGD with momentum



(a) SGD without momentum

(b) SGD with momentum



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SGD with momentum, Intuition

Figure: SGD with momentum, Intuition (CC)





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SGD with momentum

Example of SGD with momentum here.



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Adaptive Moment Estimation (Adam)

- Adapt η_t to individual parameters

$$
\begin{aligned}
m_t &= \beta_1 m_{t-1} + (1-\beta_1)\widehat{\nabla} L_t(\theta_{t-1}) \\
v_t &= \beta_2 v_{t-1} + (1-\beta_2)\widehat{\nabla} L_t(\theta_{t-1})^2
\end{aligned}
$$

- Bias correction
  (due to initialization at 0)

$$
\begin{aligned}
\hat{m}_t &= \frac{m_t}{1-\beta_1^t} \\
\hat{v}_t &= \frac{v_t}{1-\beta_2^t}
\end{aligned}
$$

- Update (squares, roots, and division act componentwise)

$$
\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon}\hat{m}_t,
$$

- Common values: $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$



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- Update (squares, roots, and division act componentwise)

$$\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon}\hat{m}_t,$$

- Common values: $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$
- RMSprop is another (similar) alternative

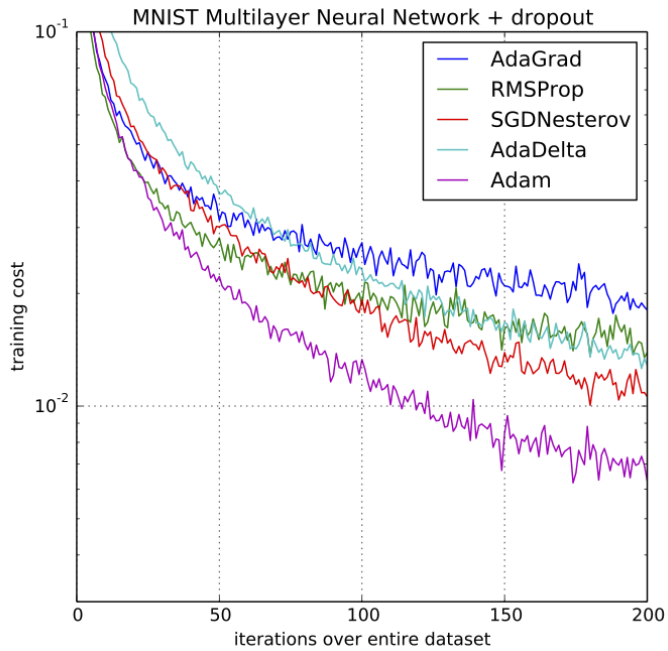


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Adam

Figure: The Adam Optimizer (Kingma and Ba, 2014)



Convergence guarantees require assumptions and depend on the variant. See:
Defossez et al (2020) "A Simple Convergence Proof of Adam and Adagrad"